



Combined biotin, folic acid, and vitamin B₁₂ supplementation given during the transition period to dairy cows: Part II. Effects on energy balance and fatty acid composition of colostrum and milk

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ABSTRACT

Biotin (B₈), folate (B₉), and vitamin B₁₂ (B₁₂) are involved in several metabolic reactions related to energy metabolism. We hypothesized that a low supply of one of these vitamins during the transition period would impair metabolic status. This study was undertaken to assess the interaction between B₈ supplement and a supplementation of B₉ and B₁₂ regarding body weight (BW) change, dry matter intake, energy balance, and fatty acid (FA) compositions of colostrum and milk fat from d -21 to 21 relative to calving. Thirty-two multiparous Holstein cows housed in tie stalls were randomly assigned, according to their previous 305-d milk yield, to 8 incomplete blocks in 4 treatments: (1) a 2-mL weekly i.m. injection of saline (0.9% NaCl; B₈-/B₉B₁₂-); (2) 20 mg/d of dietary B₈ (unprotected from ruminal degradation) and 2-mL weekly i.m. injection of 0.9% NaCl (B₈+ /B₉B₁₂-); (3) 2.6 g/d of dietary B₉ (unprotected) and 2-mL weekly i.m. injection of 10 mg of B₁₂ (B₈-/B₉B₁₂+); (4) 20 mg/d of dietary B₈, 2.6 g/d of dietary B₉, and 2-mL weekly i.m. injection of 10 mg of B₁₂ (B₈+ /B₉B₁₂+) in a 2 × 2 factorial arrangement. Colostrum was sampled at first milking, and milk samples were collected weekly on 2 consecutive milkings and analyzed for FA composition. Body condition score and BW were recorded every week throughout the trial. Within the first 21 d of lactation, B₈-/B₉B₁₂+ cows had an increased milk yield by 13.5% [45.5 (standard error, SE: 1.8) kg/d] compared with B₈-/B₉B₁₂- cows [40.1 (SE: 1.9)], whereas B₈ supplement had no effect. Even though body condition score was not affected by treatment, B₈-/B₉B₁₂+ cows had greater BW loss by 24 kg, suggesting higher mobilization of body reserves. Accordingly, milk de novo FA decreased and preformed FA concentration increased in B₈-/B₉B₁₂+ cows com-

pared with B₈-/B₉B₁₂- cows. In addition, cows in the B₈+ /B₉B₁₂- group had decreased milk de novo FA and increased preformed FA concentration compared with B₈-/B₉B₁₂- cows. Treatment had no effect on colostrum preformed FA concentration. Supplemental B₈ decreased concentrations of ruminal biohydrogenation intermediates and odd- and branched-chain FA in colostrum and milk fat. Moreover, postpartum dry matter intake for B₈+ cows tended to be lower by 1.6 kg/d. These results could indicate ruminal perturbation caused by the B₈ supplement, which was not protected from rumen degradation. Under the conditions of the current study, in contrast to B₈+ /B₉B₁₂- cows, B₈-/B₉B₁₂+ cows produced more milk without increasing dry matter intake, although these cows had greater body fat mobilization in early lactation as suggested by the FA profile and BW loss.

Key words: cyanocobalamin, folates, biotin, milk fatty acid, cattle

INTRODUCTION

It is well-established dogma that dairy cows do not need exogenous supply of B vitamins. Indeed, it has been thought that ruminal bacteria synthesize sufficient amounts of B vitamins to prevent signs of deficiency (NRC, 2001). Nevertheless, periparturient cows face challenges as major endocrinal changes occur, and the onset of lactation is often characterized by a state of negative energy balance (EB; Goff and Horst, 1997). It has been shown that plasma concentrations of folates (B₉) and vitamin B₁₂ (B₁₂) reach their nadir around calving (Girard et al., 1989; Girard and Matte, 1999), suggesting that ruminal synthesis was not sufficient to avoid fluctuations. Similarly, in the past few years, several studies have shown beneficial effects of providing either a supplement of folic acid (B₉, the synthetic form of folates) and B₁₂ (B₉B₁₂) or a supplement of biotin (B₈) to cows before calving until early lactation (Graulet et al., 2007; Preynat et al., 2009a; Chen et

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al., 2011). In some studies, B₉B₁₂ supplementation increased milk yield without modifying DMI and body fat mobilization indicators (Graulet et al., 2007; Preynat et al., 2009a), or did not change milk yield and DMI but decreased body fat mobilization indicators (Duplessis et al., 2014b, 2017b). Regarding B₈ supplementation, a meta-analysis by Chen et al. (2011) concluded that this vitamin supplementation increased milk yield and DMI.

Biotin, B₉, and B₁₂ are closely related in some metabolic pathways. Indeed, the methylated form of B₉ can give its methyl group to homocysteine to form Met. This latter reaction is B₁₂ dependent (Combs and McClung, 2017). Methionine can be used for protein synthesis or be transformed into S-adenosylmethionine, the major methyl donor in mammals (Scott, 1999). Folates and B₁₂ are also involved in purine and pyrimidine syntheses, which are DNA constituents. Moreover, before their entry into the Krebs cycle, propionate, odd-chain fatty acids (FA), and some AA need to be transformed into propionyl-CoA, then into methylmalonyl-CoA, and succinyl-CoA. The last 2 reactions require, respectively, B₈ and B₁₂ as coenzymes (Combs and McClung, 2017). Although B₉ has no known role in this pathway, some authors have reported that gene expression of methylmalonyl-CoA mutase, the enzyme that transforms methylmalonyl-CoA into succinyl-CoA with B₁₂, was greater in the presence of a combined supplement of B₉B₁₂ compared with B₁₂ alone (Preynat et al., 2010; Duplessis et al., 2017a). It is well known that, in ruminants, propionate is the main precursor for glucose synthesis (Larsen and Kristensen, 2013; Duplessis et al., 2017a). Pyruvate carboxylase requires B₈ as a coenzyme in the transformation of pyruvate into oxaloacetate, a key step of gluconeogenesis (Combs and McClung, 2017). Biotin has also a role in FA synthesis by being a coenzyme for the enzyme, allowing the elongation of acetyl-CoA into malonyl-CoA, which is the first step of FA elongation for de novo FA synthesis in the mammary gland.

It is now well recognized that cows experiencing a state of negative nutrient balance in early lactation mobilize body fat and protein to compensate for the lack of nutrients to support their high milk production. Milk FA profile changes according to EB of the animal (Bauman and Griinari, 2003). Indeed, during an episode of body fat mobilization, proportions of FA <16 C and odd-chain 5:0 to 15:0 FA decrease, whereas 16:0 and preformed FA increase, specifically *cis* (c)-9 18:1 (Stoop et al., 2009; Gross et al., 2011). Three rationales support assessing FA profile of colostrum and milk following the use of supplementary B₈ and B₉B₁₂. First, milk FA profile could be an indicator of EB and metabolic status (Dórea et al., 2017). Previous studies showed that a supplement of B₉B₁₂ changed energy

partitioning (Graulet et al., 2007; Preynat et al., 2009a; Duplessis et al., 2017b). Rosendo et al. (2004) reported that B₈ supplementation decreased lipid mobilization in postpartum cows. Second, given the role of B₈ on FA elongation (Combs and McClung, 2017), it would be of interest to assess the effect of B₈ supplementation on de novo FA in milk. Last, milk FA profile could be used as an indicator of rumen function by evaluating milk odd- and branched-chain FA and ruminal biohydrogenation intermediates (Bauman and Griinari, 2003; Fievez et al., 2012; Woolpert et al., 2017). As B₉ and B₈ were provided as dietary supplements and were not protected against ruminal degradation, it is of interest to study their potential effect on rumen function.

As these 3 vitamins have interrelated roles in metabolism, we hypothesized that the lack of one of them could impair metabolic status in early lactation. The objective of this study was to evaluate whether B₈ supplementation affects the efficiency of a supplement of B₉B₁₂ on BCS, BW, BW change, DMI, EB, and colostrum and milk FA compositions. The companion paper (Duplessis et al., 2022) presented milk performance and plasma concentrations of vitamins, energy-related metabolites, hormones, and AA.

MATERIALS AND METHODS

The Institutional Committee for Animal Care of the Sherbrooke Research and Development Center approved the experimental protocol. All procedures were conducted according to the code of practice of the National Farm Animal Care Council (2009) and the guidelines of the Canadian Council on Animal Care (2009).

Cows, Treatments, and Farm Management

A total of 32 multiparous Holstein cows from the dairy herd at the Agriculture and Agri-Food Canada Research Centre (Sherbrooke, QC, Canada) were involved in this study conducted in 2 periods. Period 1 included 17 cows [8 entering their second lactation and 9 entering their third or greater lactation (mean: 3.8, minimum: 3 and maximum: 6)] from December 2016 to June 2017; period 2 included 15 cows [6 entering their second lactation and 9 entering their third and greater lactation (mean: 3.8, min: 3 and max: 6)] from September 2017 to April 2018. They were kept in tie stalls and milked twice daily at 0730 and 1930 h under 17 h of light. Cows were assigned to 8 incomplete blocks based on their previous 305-d milk production. Within each block, they were randomly assigned to 1 of the 4 following treatments: (1) A 2-mL weekly i.m. injection of saline (0.9% NaCl; B₈–/B₉B₁₂–); (2) 20

mg/d of dietary B₈ (Sigma-Aldrich) and 2-mL weekly i.m. injection of 0.9% NaCl (**B₈+/**B₉B₁₂**–**); (3) 2.6 g/d of dietary B₉ (pteroylmonoglutamic acid, Sigma-Aldrich) and 2-mL weekly i.m. injection of 10 mg of B₁₂ (cyanocobalamin, 5,000 µg/mL, Vétoquinol; **B₈–/**B₉B₁₂**+**); (4) 20 mg/d of dietary B₈, 2.6 g/d of dietary B₉, and weekly i.m. injection of 10 mg of B₁₂ (**B₈+/**B₉B₁₂**+**). The doses of vitamins were chosen according to the trials of Graulet et al. (2007), Girard and Desrochers (2010), and Duplessis et al. (2017b). The parenteral route was chosen for B₁₂ supplementation to overcome the degradation of 80% of the dietary supplement by ruminal microorganisms (Girard et al., 2009b). Dietary supplements of vitamins were unprotected from ruminal degradation. Due to the nature of treatments (dietary or i.m. injections and color of the injected substances), administration of treatments was not blinded. The targeted treatment period was d –21 to 21 relative to parturition, but cows received treatments on average 26 [standard deviation (SD: 8) d before calving. During their previous lactation, cows produced on average 10,957 kg (SE: 105, minimum: 8,952 kg and maximum: 13,090 kg) (treatment effect, $P > 0.60$).

Cows were fed the same TMR regardless of treatments; before calving, they were fed a close-up diet twice daily distributed at 0830 and 1300 h, and switched to a lactation diet distributed once daily at 0830 h after calving. Every week, silage samples were analyzed by near-infrared reflectance spectrometry, and percentages of dietary inclusion were adjusted to keep the diet isoenergetic and isonitrogenous throughout the duration of the trial. All ingredients in the close-up and the lactation TMR were sampled every week throughout the experimental period and stored at –20°C until wet chemistry analyses. Rations and composition analysis are described in the companion paper (Duplessis et al., 2022) and Duplessis and Girard (2019). Briefly, the first 3 major ingredients in the close-up diet were corn silage (42.3% of DM), grass hay (26.9% of DM), and soybean meal (10.4% of DM; ration contained 13.7% CP, 22.8% ADF, 39.0% NDF, and 0.69 mg/kg Co on a DM basis) and those from the lactation ration were grass and legume silage (28.3% of DM), corn silage (28.2% of DM), and cracked corn (22.4% of DM; ration contained 17.1% CP, 17.4% ADF, 29.3% NDF, and 0.55 mg/kg Co on a DM basis). The quantity of TMR fed to each cow was weighed directly on the scale of the cart and recorded. Orts were weighed and recorded daily at 0800 h. Between ort removal and TMR distribution, daily dietary supplements of B₈ and B₉ mixed with 200 g of fine-ground corn were offered with a handful of corn silage. The complete diet was distributed to the animals when the vitamin supplement blend was entirely eaten. To ensure that the supplement and the

ration were eaten by the proper cows, one stall was left empty between 2 cows and separators were used to divide troughs. The quantity of TMR given during the lactation was adjusted if necessary, to allow 10% refusals. Cows had free access to water. Three days before the expected calving date, cows were transferred into a calving pen and moved into their regular stall after first colostrum milking (± 3 h following calving). Calf management has been reported elsewhere (Duplessis and Girard, 2019).

Data Collection

BW and BCS. Cows were weighed at 1315 h on 2 consecutive days every week throughout the experiment. The average BW of those 2 consecutive days was used for statistical analyses. Body condition scores were evaluated weekly by the same individual using a 1-to-5 scale with quarter points before the initiation of treatments at d –27 (SD: 8) relative to parturition and throughout the experimental period (Wildman et al., 1982; Ferguson et al., 1994).

Colostrum and Milk. Within 3.0 (SD: 2.4, minimum: 0.3, and maximum: 9.6) h following calving, colostrum samples were taken from the first milking with a milking machine in the calving pen, stored at –20°C until analyses, and colostrum yield was recorded. Milk samples were taken every week using inline milk meters from 2 consecutive milkings. One aliquot was preserved with bronopol (Brotab Milk Preservative, Systems Plus) and sent immediately to Lactanet laboratory (DHI agency, Sainte-Anne-de-Bellevue, QC, Canada) for fat percentage analysis. Another aliquot without bronopol was taken and stored at –20°C until TS and FA analyses. Milk yields were recorded at each milking.

Analyses. Total solids of colostrum were determined in duplicate with an air-forced oven according to method 990.19 (AOAC, 1989) for milk. Another aliquot of colostrum was freeze-dried and analyzed for FA. Total FA concentration of colostrum was analyzed using the 2-step methylation method developed by Kramer et al. (1997) and Garcia et al. (2014) with modifications. Fatty acids C13:0 and C21:0 were used as internal standards. A first sample incubation in 2 mL of sodium methoxide (0.5 M in methanol) at 50°C for 10 min was performed followed by a second incubation in 3 mL of HCl 10% (vol/vol) at 80°C for 10 min. Colostrum FA profile was extracted as per Hara and Radin (1978) and Gervais et al. (2005). A gas chromatograph (Agilent 7820; Agilent Technologies Canada Inc.) equipped with a HP-INNOWax capillary column (30 m $\times$ 0.32-mm i.d. $\times$ 0.25-µm film; Agilent Technologies Canada Inc.), and a flame ionization detector was used. The split ratio was 50:1. At the time of the sample injection, the

column temperature was 100°C, maintained for 1 min followed by an increase to 220°C at 5°C/min, and the temperature was then maintained for 20 min.

Milk samples were analyzed for fat percentage by mid-infrared reflectance spectrometry (MilkoScan FT 6000, Foss). Weekly milk samples from evening and morning milkings were pooled according to milk yield at each milking to obtain representative daily milk samples for FA profile analysis. Milk FA profile was analyzed as previously described by Boivin et al. (2013) with modifications. Briefly, milk sample lipid fraction was extracted and then methylated according to the method of Chouinard et al. (1997). A gas chromatograph (7890A GC; Agilent Technologies Canada Inc.) equipped with a 100-m CP-Sil 88 capillary column (0.25-mm i.d., 0.20-μm film thickness; Agilent Technologies Canada Inc.) and a flame ionization detector were used to obtain the FA composition of milk samples. At the time of sample injection, the column temperature was 80°C for 1 min, increased by 2°C/min until reaching 215°C, and then maintained for 21.5 min. Glycerol in milk fat was computed as per Schauff et al. (1992).

Calculations. Prepartum and postpartum BW changes were calculated as the absolute difference between the closest prepartum or postpartum BW data to calving (mean: 4, SD: 2, minimum: 1, and maximum: 7 absolute days relative to calving) and the closest prepartum or postpartum BW data from d 21 relative to calving (mean: 18, SD: 2, minimum: 15, and maximum: 21 absolute days relative to calving), respectively. Dry matter intake was calculated daily by subtractingorts from the quantity of TMR served and was then averaged weekly. Prepartum and postpartum EB were calculated as previously described by Duplessis et al. (2017b) based on equations of NRC (2001). Briefly, prepartum EB was based on energy supply from DMI divided by energy requirements for maintenance and pregnancy. Postpartum EB was computed using energy supply from DMI divided by energy requirements for maintenance and milk production.

Statistical Analyses

Data were analyzed according to a 2 × 2 factorial arrangement in a randomized incomplete block design. Specifically, 8, 7, 9, and 8 cows were sorted into B₈−/B₉B₁₂−, B₈+/B₉B₁₂−, B₈−/B₉B₁₂+, and B₈+/B₉B₁₂+ treatments, respectively. Two levels of B₈ (0 or 20 mg/d) and 2 levels of B₉B₁₂ (0 or 2.6 g of B₉/d and 10 mg of B₁₂/wk) were considered in the following model:

$$Y_{ijkl} = \mu + V_i + \beta_j + \delta_{ijk} + T_l + VT_{il} + \varepsilon_{ijkl},$$

in which Y_{ijkl} is the studied variable results for the k th cow of the i th treatment during l th week within the j th block, μ is the overall mean, V_i is the treatment effect, β_j is the block effect (from 1 to 8), δ_{ijk} is the error a , T_l is the week effect, VT_{il} is the interaction treatment × week, and ε_{ijkl} is the residual error. According to the factorial arrangement used, the treatment was decomposed in B₈, B₉B₁₂, and the interaction B₈ × B₉B₁₂. The block effect in the model took into account that the study was performed during 2 periods, as cows involved in the first period were not assigned to the same block number as cows in the second period. There were 6 blocks of 4 cows from each treatment, 1 block with 3 cows (1 cow missing in B₈+/B₉B₁₂−), and 1 block with 5 cows (1 extra cow in B₈−/B₉B₁₂+).

Separately for the prepartum and postpartum periods, BW, BCS, DMI, and EB were subsequently entered as dependent variables in Proc MIXED of SAS (version 9.4, SAS Institute, 2012) with repeated measures in time (weekly collection or weekly average from daily measures). Milk fat and FA concentrations were also entered as dependent variables in Proc MIXED of SAS with repeated measures in time. The following 8 covariance structures were assessed: compound symmetry, heterogeneous compound symmetry, first-order autoregressive, heterogeneous first-order autoregressive, Toeplitz, heterogeneous Toeplitz, first-order antedependence, and unstructured. For each dependent variable studied, the covariance structure leading to the smallest Bayesian information criterion was chosen. Body condition score and BW at the beginning of treatments, pre- and postpartum BW changes and colostrum TS, total FA, and FA profile were analyzed with Proc MIXED of SAS using the same model as described above except repeated measurements in time were not included. Degrees of freedom were adjusted according to the Kenward-Roger method. Residuals of all models were evaluated for the assumption of normality and homoscedasticity. Analysis of BCS at the beginning of treatment administration violated this assumption, and data were hence log-transformed. Geometric means and 95% CI were presented for this variable. For each studied variable, the INFLUENCE option in the model statement was used to detect and delete influential data as per the method of Kaps and Lamberson (2017). To simplify the text, treatment abbreviations (i.e., B₈−/B₉B₁₂−, B₈+/B₉B₁₂−, B₈−/B₉B₁₂+, and B₈+/B₉B₁₂+) or individual supplement abbreviations (i.e., B₈−, B₈+, B₉B₁₂− and B₉B₁₂+) will be used in the following sections. Statistical comparisons were the main B₈ effect (B₈−/B₉B₁₂− plus B₈−/B₉B₁₂+ vs. B₈+/B₉B₁₂− plus B₈+/B₉B₁₂), the main B₉B₁₂ effect (B₈−/B₉B₁₂− plus B₈−/B₉B₁₂− vs. B₈−/B₉B₁₂+ plus B₈+/B₉B₁₂+), and B₈ × B₉B₁₂ and the B₈ × B₉B₁₂ interactions. When an

interaction was significant, the SLICE option of the LSMEANS statement was used to help its interpretation, allowing us to compare the simple effects of B₈ or B₉B₁₂ within each level of the other treatment; that is, for each level of B₈ supplementation (B₈– and B₈+) comparing the response to the B₉B₁₂ supplement or inversely, comparing the response to the B₈ supplement within each level of B₉B₁₂ supplement (B₉B₁₂– and B₉B₁₂+), and *P*-values of each simple effect are reported in the Results section. Results were considered significant if *P*-value ≤ 0.05 and a tendency if 0.05 < *P* < 0.10. Blocks were incomplete due to cow availability at the barn and because 2 cows in B₈+/B₉B₁₂– treatment were culled at calving due to locomotion and disease problems unrelated to treatment.

RESULTS

Pretreatment Animal Characteristics

Body condition score did not differ among treatments before enrollment: 27 (SD: 8) d before calving [geometric mean (95% CI): 2.74 (2.46–3.05) for B₈–/B₉B₁₂–, 2.99 (2.66–3.36) for B₈+/B₉B₁₂–, 2.99 (2.70–3.32) for B₈–/B₉B₁₂+, and 3.15 (2.83–3.51) for B₈+/B₉B₁₂+; *P* > 0.19]. The first BW was taken on average 22 ± 8 d before calving and was higher by 57 kg for B₈+ cows compared with B₈– cows (B₈ effect, *P* = 0.02). No difference in first BW measurement was noted for B₉B₁₂+ and B₉B₁₂– cows (B₉B₁₂ effect, *P* = 0.67). Considering the result on BW for the B₈ supplement, pretreatment BW was considered a covariate in parturition and postpartum models (*P* < 0.0001).

BW, BCS, DMI, and EB

Prepartum. Cows in B₈+/B₉B₁₂+ group tended to have a lower BW than cows in B₈–/B₉B₁₂+ group (SLICE test, *P* = 0.08), whereas no effect was noted for B₉B₁₂– cows (SLICE test, *P* = 0.44; B₈ × B₉B₁₂ interaction, *P* = 0.06; Table 1). No difference in BW change, BCS, or DMI was observed among treatments (*P* ≥ 0.19). Biotin supplement decreased EB by 5.5% regardless of B₉B₁₂ supplementation (B₈ effect, *P* = 0.04). Energy balance as a % of requirement of B₉B₁₂+ cows decreased from 123 to 119% during the 21 d before parturition (SLICE test, *P* < 0.0001), whereas no such effect was observed for B₉B₁₂– cows (SLICE test, *P* = 0.13; B₉B₁₂ × week interaction, *P* = 0.008; Figure 1).

Postpartum. Despite the fact that no significant effect of treatment was observed on BW or BCS during the postpartum period (*P* > 0.16), BW loss was significantly greater for B₈–/B₉B₁₂+ cows than for B₈–/B₉B₁₂– cows (SLICE test, *P* = 0.01), whereas no effect

was observed when supplementary B₈ was fed (SLICE test, *P* = 0.75; B₈ × B₉B₁₂ interaction, *P* = 0.05). Supplemental B₈ tended to decrease DMI by 1.6 kg/d (B₈ effect, *P* = 0.09; Table 1). Although we observed a significant B₈ × B₉B₁₂ interaction on EB (*P* = 0.05), no clear effect of B₈ within each level of B₉B₁₂ or vice versa (SLICE test, *P* > 0.15; Figure 1) was detectable.

Colostrum FA Profile

Colostrum yield and TS did not differ among treatments (*P* > 0.32) and averaged 4.2 ± 2.2 kg and 24.8 ± 1.2%, respectively. Percentage of total FA on a DM basis was lower by 5.5 percentage points for B₉B₁₂+ cows compared with B₉B₁₂– cows (B₉B₁₂ effect, *P* = 0.04; Table 2). Cows in the B₈+/B₉B₁₂– group had a decreased colostrum fat concentration of ruminal biohydrogenation intermediates, such as *trans* (t)11 18:1, t11c15 18:2, and c9t11 18:2, compared with the B₈–/B₉B₁₂– group (SLICE test, *P* < 0.02), but no effect was noted with B₉B₁₂ supplementation (SLICE test, *P* > 0.63; B₈ × B₉B₁₂ interaction, *P* ≤ 0.08). De novo FA concentration in colostrum fat was greater for B₈–/B₉B₁₂+ cows than for B₈+/B₉B₁₂+ cows (SLICE test, *P* = 0.04), mainly due to the increased concentration of 14:0, whereas no effect was detectable for B₉B₁₂– cows (SLICE test, *P* = 0.41; B₈ × B₉B₁₂ interaction, *P* = 0.04, Table 2). Biotin supplement increased, or tended to, colostrum fat concentrations of some n-6 FA; namely, c6,9,12 18:3; c8,11,14 20:3; and c5,8,11,14 20:4 in colostrum (B₈ effect, *P* ≤ 0.06).

Milk FA Profile

As shown in Duplessis et al. (2022), milk yield within the first 3 wk of lactation was affected by vitamin supplementation (B₈ × B₉B₁₂ interaction, *P* = 0.06). Indeed, B₈–/B₉B₁₂+ cows produced more milk than B₈–/B₉B₁₂– cows [B₈–/B₉B₁₂–: 40.1 (95% CI: 36.2–44.0); B₈+/B₉B₁₂–: 43.3 (39.0–47.5); B₈–/B₉B₁₂+: 45.5 (41.8–49.2); B₈+/B₉B₁₂+: 41.0 (37.1–44.9) kg/d]. Neither milk fat percentages (*P* > 0.37) nor milk fat yields (*P* > 0.33) differed according to treatment. Milk fat percentage averaged 4.53, 4.35, 4.26, and 4.58 ± 0.29%, whereas milk fat yields averaged 1.83, 1.84, 2.05, and 1.85 ± 0.13 kg/d for B₈–/B₉B₁₂–, B₈+/B₉B₁₂–, B₈–/B₉B₁₂+, and B₈+/B₉B₁₂+ groups, respectively.

Significant B₈ × B₉B₁₂ interactions or tendencies (*P* ≤ 0.06) occurred for individual milk de novo FA concentrations, such as 6:0, 8:0, 10:0, and 12:0; nevertheless, no clear effect of each main factor within each simple level was detectable (Table 3). However, B₈–/B₉B₁₂+ cows had a decreased milk fat concentration of 14:0 compared with B₈–/B₉B₁₂– cows (SLICE test,

Table 1. Body weight, BW change, BCS, DMI, and energy balance (EB) of cows receiving supplementary biotin, folic acid, and vitamin B₁₂, before and after parturition¹

Item	Treatment ²				SEM	P-value ³		
	B ₈ -/B ₉ B ₁₂ -	B ₈ + /B ₉ B ₁₂ -	B ₈ -/B ₉ B ₁₂ +	B ₈ + /B ₉ B ₁₂ +		B ₈	B ₉ B ₁₂	B ₈ × B ₉ B ₁₂
Cows, n	8	7	9	8				
BW, kg								
Pregalving ⁴	768	776	783	764	8	0.53	0.81	0.06
Postcalving ⁴	705	705	720	690	12	0.22	0.99	0.16
BW change, kg								
Pregalving	21.5	21.4	26.1	18.3	4.8	0.35	0.86	0.36
Postcalving	-14.5	-25.5	-38.3	-22.5	6.9	0.71	0.12	0.05
BCS ⁵								
Pregalving ⁴	2.99	3.14	3.02	3.02	0.07	0.25	0.51	0.23
Postcalving ⁴	2.77	2.85	2.74	2.79	0.07	0.34	0.53	0.82
DMI, kg/d								
Pregalving ⁴	13.9	13.8	14.0	13.7	0.1	0.19	0.65	0.51
Postcalving ^{4,6}	22.0	20.1	21.9	20.4	1.0	0.09	0.91	0.84
EB, % of requirement								
Pregalving ^{4,6,7}	125	118	123	119	3	0.04	0.99	0.54
Postcalving ⁴	72	65	65	73	4	0.96	0.93	0.05

¹Results before parturition included data from -21 to -1 DIM; those after calving included data from 1 to 21 DIM.

²Treatments are defined as follows: B₈-/B₉B₁₂- = 2-mL weekly i.m. injection of saline (0.9% NaCl); B₈+ /B₉B₁₂- = 20 mg/d of dietary biotin and 2-mL weekly i.m. injection of 0.9% NaCl; B₈-/B₉B₁₂ + = 2.6 g/d of dietary folic acid and 2-mL weekly i.m. injection of 10 mg of vitamin B₁₂; B₈+ /B₉B₁₂ + = 20 mg/d of dietary biotin, 2.6 g/d of dietary folic acid, and 2-mL weekly i.m. injection of 10 mg/wk of vitamin B₁₂. Treatments were given from d -26 to 21 relative to parturition. Least squares means and SEM are presented.

³Statistical comparisons: B₈ effect (B₈-/B₉B₁₂- plus B₈-/B₉B₁₂ + vs. B₈+ /B₉B₁₂- plus B₈+ /B₉B₁₂ +); B₉B₁₂ effect (B₈-/B₉B₁₂- plus B₈-/B₉B₁₂ + vs. B₈+ /B₉B₁₂- plus B₈+ /B₉B₁₂ +); B₈ × B₉B₁₂ (the interaction effect between each simple effect levels of B₈ and B₉B₁₂; that is, B₈- = B₈-/B₉B₁₂- vs. B₈-/B₉B₁₂ +; B₈+ = B₈+ /B₉B₁₂- vs. B₈+ /B₉B₁₂ +; B₉B₁₂- = B₈-/B₉B₁₂- vs. B₈+ /B₉B₁₂-; B₉B₁₂ + = B₈-/B₉B₁₂ + vs. B₈+ /B₉B₁₂ +).

⁴Week effect, $P < 0.02$.

⁵According to a 1-to-5 scale with quarter points (Wildman et al., 1982; Ferguson et al., 1994).

⁶B₈ × B₉B₁₂ × week interaction, $P < 0.08$.

⁷B₉B₁₂ × week interaction, $P = 0.008$.

$P = 0.04$), whereas no effect was noted for B₈+ cows (SLICE test, $P = 0.11$; B₈ × B₉B₁₂ interaction, $P = 0.01$). Similar results were obtained for the sum of de novo FA concentration (B₈ × B₉B₁₂ interaction, $P = 0.03$). Concentrations of *iso* 13:0, *iso* 14:0, and *iso* 15:0 were lower for B₈+ /B₉B₁₂- cows than for B₈-/B₉B₁₂- cows (SLICE test, $P < 0.02$), whereas no effect was detected for B₉B₁₂+ cows (SLICE test, $P > 0.24$; B₈ × B₉B₁₂ interaction, $P < 0.06$, Table 3). Dietary B₈ supplement decreased concentrations of some intermediates of ruminal biohydrogenation in milk fat (i.e., *t5* 18:1, *t6*-8 18:1, *t11* 18:1, *t11c15* 18:2, and *c9t11* 18:2; B₈ effect, $P \leq 0.08$, Table 3). For 16:0 and other intermediates of ruminal biohydrogenation (i.e., *t9* 16:1, *t4* 18:1, *t11* 18:1, *t11c15* 18:2), significant B₈ × B₉B₁₂ interactions were obtained ($P \leq 0.04$), implying that B₈+ /B₉B₁₂- group had lower milk fat in these FA compared with B₈-/B₉B₁₂- group (SLICE test, $P < 0.001$), although no effect was detectable for B₉B₁₂+ groups (SLICE test, $P > 0.37$). Milk fat concentration of *c9* 18:1 and the sum of preformed FA concentration in milk fat were increased by 11.6% for B₈-/B₉B₁₂+ group compared with B₈-/B₉B₁₂- cows (SLICE test,

$P = 0.02$), although no effect occurred when supplementary B₈ was fed (SLICE test, $P > 0.17$; B₈ × B₉B₁₂ interaction, $P < 0.02$, Table 3). Opposite results were obtained for glycerol concentration. Milk FA yields are presented in Supplemental Table S1 (<https://data.mendeley.com/datasets/3p94t45zst/2>; Duplessis et al., 2022).

DISCUSSION

Milk FA profile can be a noninvasive proxy for rumen function (Bauman and Griinari, 2003; Vlaeminck et al., 2006). It has been shown that unprotected dietary B₉ supplement did not affect rumen fermentation characteristics (Girard et al., 2009a), but that unprotected dietary B₈ supplement affected nitrogen metabolism in the rumen (Girard and Desrochers, 2010), suggesting a modification in rumen function with the dietary B₈ supplement. To our knowledge, this is the first time that colostrum and milk FA profiles have been reported following a combined supplementation of B₈ and B₉B₁₂. In a previous study, 20 mg/d of unprotected dietary B₈ supplement provided during the peripartum period

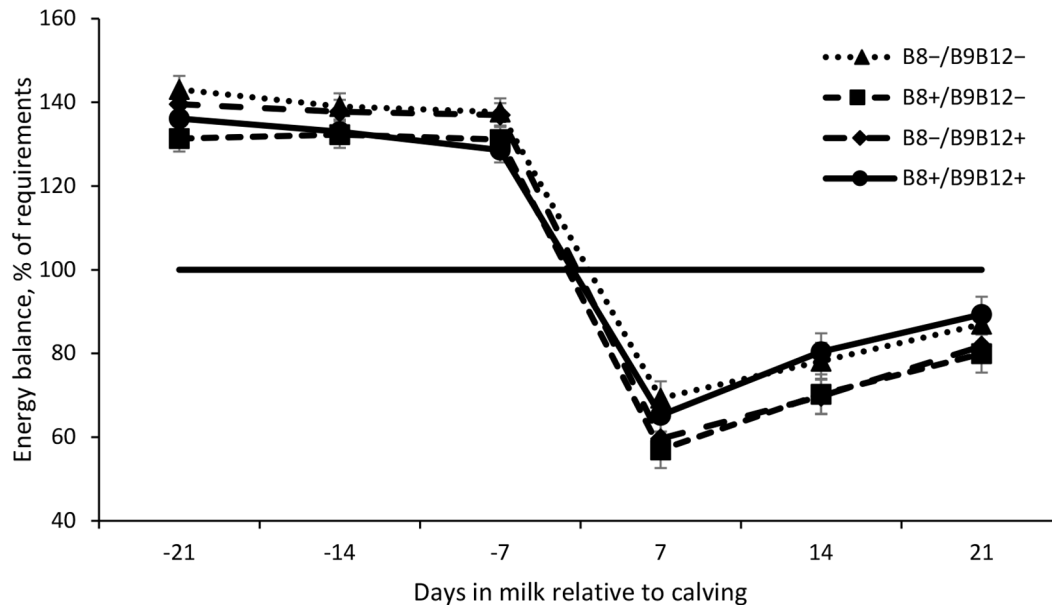


Figure 1. Energy balance of periparturient Holstein cows receiving either a 2-mL weekly i.m. injection of saline 0.9% NaCl (B₈-/B₉B₁₂-), 20 mg/d of dietary biotin and 2-mL weekly i.m. injection of saline 0.9% NaCl (B₈+/B₉B₁₂-), 2.6 g/d of dietary folic acid and 2-mL weekly i.m. injection of 10 mg of vitamin B₁₂ (B₈-/B₉B₁₂+), or 20 mg/d of dietary biotin, 2.6 g/d of dietary folic acid, and 2-mL weekly i.m. injection of 10 mg/wk of vitamin B₁₂ (B₈+/B₉B₁₂+). Treatments were given from d -26 to 21 relative to parturition. Error bars represent SE. Before calving: B₉B₁₂ × week interaction, $P = 0.008$; after calving: B₈ × B₉B₁₂ interaction, $P = 0.05$.

decreased ruminal biohydrogenation intermediates of milk fat (Enjalbert et al., 2008), which is in agreement with our results, where these FA are decreased in both colostrum and milk fat by B₈ supplementation. Intermediates of ruminal biohydrogenation are produced during the transformation process of dietary UFA to SFA by ruminal bacteria (Jenkins et al., 2008) and end up in milk. This is why their concentrations can be used as a proxy of rumen function (Harvatine et al., 2009). *trans*-11 18:1 and *c9t11* 18:2 are the most important intermediates of biohydrogenation of linoleic acid (i.e., *c9,12* 18:2), which is abundant in grass (Harvatine et al., 2009). Cows in the B₈+/B₉B₁₂- group had decreased colostrum and milk concentrations of these intermediates, suggesting an alteration in the ruminal microbial populations (Harvatine et al., 2009). Additionally, milk fat concentrations of odd- and branched-chain FA were altered by B₈ supplementation. These FA originate from different primers of ruminal bacterial fermentation (Vlaeminck et al., 2006). Odd-chain FA (e.g., 15:0 and 17:0) have propionate or valerate as primers, whereas branched-chain FA (e.g., *anteiso* 13:0 and *iso* 15:0) have branched-chain AA as precursors and are synthesized by rumen bacteria (Fievez et al., 2012). It has been shown that milk odd- and branched-chain FA profile can be a noninvasive indicator of bacterial population shift in the rumen and ruminal fermentation (Fievez et al., 2012). Our current

results could therefore indicate altered rumen function caused by a bacterial population shift (Harvatine et al., 2009; Fievez et al., 2012) when cows are fed unprotected dietary B₈. This rumen perturbation could explain why no increase in milk yield was observed for B₈+/B₉B₁₂+ cows compared with B₈-/B₉B₁₂+ cows (Duplessis et al., 2022). The current experimental design did not allow further evaluation of the mechanism by which unprotected dietary B₈ supplementation altered rumen function. A possible explanation might be an effect of the supplement on the ruminal microflora. It has been shown that some cellulolytic ruminal bacterial strains have a B₈ requirement to grow (Scott and Dehority, 1965), whereas some strains can synthesize it (Al-Dilaimi et al., 2014). Thus, current results suggest that B₈ supplementation might have favored some bacterial strains at the expense of others, thus creating a different microbial equilibrium, although the rumen microbiota is a complex environment with high resiliency following a perturbation (Weimer, 2015). For this reason, more studies are needed to evaluate a possible mechanism. Moreover, DMI tended to be lower for cows receiving dietary B₈ supplementation. From the current assessment, it is not possible to determine whether the possible rumen perturbation was responsible for the decreased DMI observed with dietary B₈ supplement, or, alternatively, whether the modification in milk FA composition was a consequence of a reduced DMI. To our

Table 2. Colostrum fat composition (g/100 g of fat) of cows receiving supplements of biotin, folic acid, and vitamin B₁₂

Item	Treatment ¹				SEM	P-value ²		
	B ₈ -/B ₉ B ₁₂ -	B ₈ + /B ₉ B ₁₂ -	B ₈ -/B ₉ B ₁₂ +	B ₈ + /B ₉ B ₁₂ +		B ₈	B ₉ B ₁₂	B ₈ × B ₉ B ₁₂
Cows, n	8	7	9	8				
Total FA, ³ % of DM	22.643	24.720	19.277	17.192	2.474	1.00	0.04	0.40
4:0	1.674	1.815	1.761	1.787	0.120	0.49	0.81	0.64
6:0	0.872	0.905	0.914	0.912	0.050	0.74	0.61	0.71
8:0	0.465	0.471	0.489	0.491	0.025	0.86	0.38	0.93
10:0	1.152	1.111	1.184	1.179	0.053	0.66	0.35	0.74
c9 10:1	0.063	0.057	0.064	0.065	0.006	0.70	0.42	0.56
11:0	0.023	0.021	0.018	0.021	0.001	0.74	0.12	0.08
12:0	2.036	2.092	2.180	2.069	0.067	0.68	0.37	0.22
c9 12:1	0.043	0.040	0.045	0.041	0.003	0.29	0.62	0.91
iso 13:0	0.017	0.015	0.016	0.018	0.002	0.89	0.44	0.33
anteiso 13:0	0.012	0.011	0.010	0.013	0.001	0.54	0.98	0.20
iso 14:0	0.046	0.040	0.044	0.050	0.003	0.94	0.29	0.09
14:0	11.705	12.156	12.338	11.380	0.327	0.44	0.83	0.04
c9 14:1	1.133	1.100	1.127	0.969	0.081	0.23	0.41	0.45
c11 14:1	0.023	0.022	0.021	0.020	0.002	0.65	0.34	0.83
iso 15:0	0.149	0.135	0.153	0.155	0.011	0.60	0.28	0.44
anteiso 15:0	0.230	0.227	0.236	0.211	0.020	0.50	0.82	0.58
15:0	0.733	0.749	0.761	0.757	0.029	0.84	0.54	0.73
iso 16:0	0.182	0.160	0.181	0.194	0.010	0.69	0.12	0.09
16:0	37.650	36.952	37.105	35.850	1.030	0.35	0.43	0.79
t9 16:1	0.086	0.071	0.079	0.079	0.006	0.17	0.97	0.21
c9 16:1	2.489	2.295	2.374	2.391	0.166	0.59	0.95	0.53
c11 16:1	0.107	0.118	0.110	0.106	0.008	0.69	0.55	0.35
c13 16:1	0.108	0.104	0.112	0.102	0.008	0.40	0.85	0.62
iso 17:0 + t10 16:1	0.264	0.255	0.275	0.271	0.015	0.65	0.36	0.87
anteiso 17:0 + c10 16:1	0.424	0.425	0.433	0.423	0.021	0.84	0.87	0.80
17:0	0.261	0.270	0.271	0.271	0.016	0.77	0.69	0.77
c7 17:1	0.156	0.157	0.156	0.159	0.008	0.84	0.89	0.89
iso 18:0	0.055	0.052	0.057	0.055	0.003	0.26	0.34	0.92
18:0	4.739	4.720	4.583	5.111	0.334	0.45	0.72	0.42
t4 18:1	0.015	0.011	0.011	0.012	0.001	0.37	0.21	0.17
t5 18:1	0.013	0.009	0.009	0.010	0.001	0.02	0.03	0.008
t6-8 18:1	0.182	0.158	0.163	0.161	0.010	0.21	0.44	0.29
t9 18:1	0.142	0.140	0.140	0.124	0.009	0.32	0.27	0.42
t10 18:1	0.146	0.144	0.148	0.145	0.011	0.81	0.89	0.94
t11 18:1	0.628	0.467	0.508	0.540	0.045	0.15	0.61	0.04
t12 18:1	0.238	0.239	0.254	0.240	0.016	0.68	0.58	0.65
t15 18:1	0.110	0.099	0.110	0.119	0.009	0.89	0.25	0.28
t16 18:1	0.131	0.123	0.133	0.154	0.012	0.55	0.14	0.18
c9 18:1	15.340	15.734	15.289	16.727	0.684	0.19	0.49	0.45
c11 18:1	0.602	0.602	0.551	0.635	0.032	0.20	0.79	0.20
c12 18:1	0.120	0.130	0.145	0.156	0.011	0.36	0.03	0.96
c13 18:1	0.076	0.083	0.074	0.084	0.007	0.22	0.92	0.86
c15 18:1	0.028	0.027	0.027	0.033	0.002	0.15	0.13	0.07
c9t12 18:2	0.036	0.035	0.039	0.039	0.003	0.84	0.21	0.89
t9c12 18:2	0.033	0.032	0.034	0.031	0.002	0.46	0.92	0.55
t9t12 18:2	0.012	0.012	0.010	0.009	0.001	0.66	0.02	0.79
c9t13 18:2	0.203	0.214	0.220	0.260	0.013	0.06	0.02	0.28
t8c13 18:2	0.052	0.048	0.053	0.053	0.003	0.54	0.34	0.57
t11c15 18:2	0.057	0.043	0.049	0.049	0.004	0.11	0.81	0.08
c9,12 18:2	1.824	1.848	1.762	1.945	0.082	0.21	0.83	0.33
c9t11 18:2	0.276	0.207	0.235	0.240	0.015	0.04	0.80	0.02
t10c12 18:2	0.008	0.007	0.008	0.007	0.001	0.23	0.92	0.94
c6,9,12 18:3	0.041	0.050	0.046	0.048	0.003	0.06	0.75	0.18
c9,12,15 18:3	0.470	0.462	0.447	0.501	0.021	0.28	0.68	0.14
c9t11c15 18:3	0.026	0.024	0.023	0.025	0.002	0.75	0.55	0.26
c6,9,12,15 18:4	0.025	0.029	0.028	0.032	0.003	0.20	0.38	0.91
19:0	0.075	0.072	0.074	0.075	0.007	0.85	0.87	0.75
20:0	0.112	0.110	0.108	0.122	0.007	0.40	0.57	0.24
c9 20:1	0.023	0.024	0.025	0.026	0.001	0.49	0.20	0.72
c11,14 20:2	0.037	0.029	0.031	0.035	0.002	0.52	0.97	0.008
c8,11,14 20:3	0.198	0.254	0.223	0.267	0.025	0.06	0.45	0.81
c5,8,11,14 20:4	0.277	0.342	0.312	0.340	0.017	0.01	0.34	0.29

Continued

Table 2 (Continued). Colostrum fat composition (g/100 g of fat) of cows receiving supplements of biotin, folic acid, and vitamin B₁₂

Item	Treatment ¹				SEM	<i>P</i> -value ²		
	B ₈ -/B ₉ B ₁₂ -	B ₈ + /B ₉ B ₁₂ -	B ₈ -/B ₉ B ₁₂ +	B ₈ + /B ₉ B ₁₂ +		B ₈	B ₉ B ₁₂	B ₈ × B ₉ B ₁₂
c8,11,14,17 20:4	0.017	0.017	0.015	0.018	0.002	0.22	0.68	0.47
c5,8,11,14,17 20:5	0.105	0.136	0.122	0.121	0.010	0.14	0.94	0.11
22:0	0.053	0.053	0.049	0.057	0.004	0.39	0.96	0.38
c13,16 22:2	0.012	0.011	0.011	0.014	0.002	0.57	0.47	0.19
c7,10,13,16 22:4	0.040	0.044	0.042	0.045	0.003	0.21	0.55	0.86
c7,10,13,16,19 22:5	0.190	0.221	0.202	0.213	0.018	0.25	0.92	0.59
c4,7,10,13,16,19 22:6	0.033	0.033	0.032	0.033	0.003	0.98	0.96	0.90
24:0	0.035	0.032	0.030	0.036	0.003	0.62	0.68	0.12
Glycerol	11.066	11.103	11.111	11.073	0.033	1.00	0.82	0.27
Sum								
De novo FA ⁴	17.491	17.950	18.361	17.125	0.398	0.33	0.96	0.04
C16	40.439	39.541	39.781	38.526	1.172	0.36	0.48	0.88
Preformed FA ⁵	29.340	29.603	29.003	31.498	1.178	0.25	0.51	0.35

¹Treatments are defined as follows: B₈-/B₉B₁₂- = 2-mL weekly i.m. injection of saline 0.9% NaCl; B₈+/B₉B₁₂- = 20 mg/d of dietary biotin and 2-mL weekly i.m. injection of saline 0.9% NaCl; B₈-/B₉B₁₂+ = 2.6 g/d of dietary folic acid and 2-mL weekly i.m. injection of 10 mg of vitamin B₁₂; B₈+/B₉B₁₂+ = 20 mg/d of dietary biotin, 2.6 g/d of dietary folic acid, and 2-mL weekly i.m. injection of 10 mg/wk of vitamin B₁₂. Treatments were given from d -26 to 21 relative to parturition. Least squares means and SEM are presented.

²Statistical comparisons: B₈ effect (B₈−/B₉B₁₂− plus B₈−/B₉B₁₂+ vs. B₈+/B₉B₁₂− plus B₈+/B₉B₁₂+); B₉B₁₂ effect (B₈−/B₉B₁₂− plus B₈−/B₉B₁₂+ vs. B₈−/B₉B₁₂+ plus B₈+/B₉B₁₂+); B₈ × B₉B₁₂ (the interaction effect between each simple effect levels of B₈ and B₉B₁₂; that is, B₈− = B₈−/B₉B₁₂− vs. B₈−/B₉B₁₂+; B₈+ = B₈+/B₉B₁₂− vs. B₈+/B₉B₁₂+; B₉B₁₂− = B₈−/B₉B₁₂− vs. B₈+/B₉B₁₂−; B₉B₁₂+ = B₈−/B₉B₁₂+ vs. B₈+/B₉B₁₂+).

 $^3c = cis; t = trans.$ ⁴Sum of straight even-chain fatty acids (FA) from C6 to C14.

⁵Sum of branched-chain FA (*iso* 13:0, *iso* 14:0, *iso* 15:0, *anteiso* 15:0, *iso* 16:0, *iso* 17:0, and *anteiso* 17:0), odd-chain FA (13:0 and 15:0), and all FA with a chain length greater than 17 C.

knowledge, no other research has reported that DMI decreased with B₈ supplementation. Conversely, in a meta-analysis, Chen et al. (2011) concluded that DMI increased by about 3.9% following B₈ supplementation. Weerathilake et al. (2019) presented the milk FA profile of early lactating cows receiving i.m. injections of B₁₂, but did not report any significant differences compared with control cows. However, the current study differed in the fact that B₁₂ supplement was always combined with B₉ and that milk and TS yields were increased in B₈–/B₉B₁₂+ cows.

Milk FA profile can also be used as a noninvasive indicators of EB and metabolic status in early lactation (Pires et al., 2022). Despite no treatment effect on postpartum EB, B₈−/B₉B₁₂+ cows lost more BW after calving, which suggested that they mobilized more body fat reserve than did B₈−/B₉B₁₂− cows. As a consequence, they produced milk fat with a greater percentage of preformed FA and a lower percentage of de novo FA. It has been shown that body fat mobilization results in a mammary uptake of long-chain FA (i.e., C >16) originating from adipose tissue and, therefore, increases their presence in milk fat (Palmquist et al., 1993; Stoop et al., 2009; Gross et al., 2011). Moreover, long-chain FA inhibit mammary de novo FA synthesis (i.e., C <16), and this inhibition is greater as the number of carbons increases from C6 to C16, due to a concomi-

tant increase in acetyl unit requirement (Palmquist et al., 1993). This could explain why $B_8-/B_9B_{12}+$ cows had decreased milk fat concentrations of 14:0 and 16:0, but not for shorter-chain de novo FA. Supplementation in B_9B_{12} increased milk fat concentration of *c*9 18:1, the major FA found in adipose tissue of dairy cows (Rukkwamsuk et al., 2000), and its milk fat concentration is an indicator for lipomobilization (Pires et al., 2022). This is in line with the higher plasma concentration in NEFA of these cows (Duplessis et al., 2022) and with Jorjong et al. (2014), who observed that cows with greater milk *c*9 18:1 were more likely to have greater plasma NEFA concentration. The second main FA contained in bovine adipose tissue is 16:0 (Christie, 1981). However, a tendency for lower milk fat concentration of 16:0 in $B_8-/B_9B_{12}+$ group was observed. Because 16:0 is partly de novo synthesized in the mammary gland, the effects of negative EB on the concentration of this FA in milk fat are not consistent. Stoop et al. (2009) reported that cows in negative EB, mobilizing more fat reserve, had greater 16:0 concentrations in their milk, whereas Gross et al. (2011) found the opposite. In spite of greater BW loss and milk FA profile suggesting higher body fat mobilization for $B_8-/B_9B_{12}+$ cows, these cows produced, respectively, 13.5% and 13.9% higher milk and TS yields combined with greater feed efficiency, as suggested by a 52.5% lower residual feed intake

Table 3. Milk fat composition (g/100 g of fat) within the first 21 d of lactation of cows receiving supplements of biotin, folic acid, and vitamin B₁₂

Item	Treatment ¹				SEM	P-value ²		
	B ₈ -/B ₉ B ₁₂ -	B ₈ + /B ₉ B ₁₂ -	B ₈ -/B ₉ B ₁₂ +	B ₈ + /B ₉ B ₁₂ +		B ₈	B ₉ B ₁₂	B ₈ × B ₉ B ₁₂
Cows, n	8	7	9	8				
Fatty acid ³								
4:0 ^{4,5}	3.155	3.140	3.059	3.110	0.103	0.85	0.52	0.73
6:0 ^{4,5}	1.690	1.539	1.552	1.686	0.078	0.90	0.95	0.06
8:0 ^{4,5,6}	0.931	0.824	0.837	0.960	0.061	0.88	0.71	0.05
10:0 ^{4,5,6}	2.073	1.716	1.794	2.121	0.174	0.93	0.70	0.05
c9 10:1 ^{4,5}	0.150	0.128	0.131	0.144	0.014	0.76	0.90	0.19
11:0 ^{4,7}	0.045	0.038	0.054	0.042	0.009	0.25	0.42	0.72
12:0 ^{4,5,6}	2.400	1.961	2.040	2.403	0.208	0.85	0.83	0.05
c9 12:1 ⁴	0.046	0.037	0.040	0.046	0.006	0.80	0.80	0.15
iso 13:0 ⁴	0.023	0.018	0.021	0.022	0.002	0.14	0.40	0.06
anteiso 13:0 ⁴	0.024	0.022	0.022	0.025	0.002	0.98	0.89	0.30
iso 14:0 ⁴	0.062	0.045	0.047	0.054	0.004	0.19	0.49	0.007
14:0 ^{4,6}	9.244	7.633	7.952	8.679	0.464	0.32	0.78	0.01
c9 14:1 ⁴	0.552	0.603	0.545	0.560	0.037	0.38	0.51	0.64
c11 14:1 ⁴	0.027	0.024	0.024	0.028	0.003	0.73	0.86	0.21
iso 15:0 ⁴	0.158	0.125	0.142	0.143	0.008	0.04	0.87	0.04
anteiso 15:0 ^{4,6}	0.348	0.333	0.346	0.314	0.025	0.33	0.66	0.70
15:0 ⁴	0.927	0.767	0.950	0.788	0.077	0.03	0.76	0.99
iso 16:0	0.187	0.173	0.181	0.188	0.015	0.81	0.77	0.47
16:0 ⁴	27.423	24.527	24.949	24.731	0.662	0.02	0.08	0.04
t9 16:1 ^{4,7}	0.054	0.047	0.047	0.049	0.003	0.06	0.70	0.008
c9 16:1 ⁴	1.365	1.890	1.662	1.646	0.133	0.05	0.83	0.04
c11 16:1 ⁴	0.055	0.100	0.071	0.071	0.009	0.02	0.52	0.02
c13 16:1 ⁴	0.099	0.091	0.089	0.100	0.10	0.90	0.97	0.29
iso 17:0 + t10 16:1 ⁴	0.290	0.291	0.317	0.295	0.010	0.32	0.16	0.30
anteiso 17:0 + c10 16:1 ⁴	0.437	0.489	0.500	0.434	0.018	0.75	0.88	0.003
17:0 ^{4,6}	0.319	0.350	0.355	0.314	0.014	0.72	0.98	0.01
c7 17:1 ⁴	0.149	0.208	0.192	0.172	0.014	0.15	0.76	0.005
iso 18:0 ⁴	0.058	0.061	0.066	0.061	0.004	0.74	0.25	0.22
18:0 ⁴	8.505	8.940	8.971	8.870	0.458	0.70	0.64	0.54
t4 18:1 ^{4,7}	0.019	0.016	0.017	0.018	0.001	0.10	0.99	0.01
t5 18:1 ^{4,5,7}	0.019	0.017	0.018	0.017	0.001	0.08	0.94	0.46
t6-8 18:1 ⁴	0.265	0.235	0.260	0.241	0.010	0.02	0.97	0.59
t9 18:1 ⁴	0.226	0.232	0.240	0.213	0.011	0.31	0.82	0.12
t10 18:1 ⁴	0.303	0.316	0.369	0.272	0.031	0.18	0.72	0.09
t11 18:1	1.050	0.681	0.865	0.875	0.068	0.009	0.95	0.007
t12 18:1	0.351	0.320	0.352	0.336	0.016	0.14	0.60	0.63
t15 18:1 ⁴	0.257	0.250	0.270	0.255	0.013	0.37	0.45	0.72
t16 18:1	0.302	0.287	0.315	0.300	0.014	0.24	0.30	0.99
c9 18:1 ⁴	19.580	24.785	23.415	22.448	1.198	0.07	0.51	0.01
c11 18:1 ⁴	0.718	0.896	0.890	0.781	0.042	0.38	0.46	0.001
c12 18:1 ⁵	0.244	0.241	0.249	0.251	0.016	0.96	0.65	0.88
c13 18:1 ⁴	0.099	0.182	0.127	0.131	0.02	0.008	0.43	0.02
c15 18:1	0.064	0.067	0.068	0.064	0.004	0.92	0.85	0.32
c9t12 18:2 ⁶	0.061	0.057	0.061	0.060	0.003	0.34	0.60	0.56
t9c12 18:2	0.035	0.036	0.038	0.036	0.002	0.68	0.51	0.54
t9t12 18:2	0.013	0.013	0.013	0.013	0.001	0.73	0.74	0.89
c9t13 18:2 ⁴	0.257	0.255	0.292	0.290	0.010	0.82	0.002	0.98
t8c13 18:2 ⁴	0.061	0.070	0.067	0.062	0.005	0.80	0.82	0.15
t11c15 18:2 ^{4,6}	0.085	0.062	0.076	0.071	0.004	0.003	0.89	0.04
c9,12 18:2 ^{4,5}	1.865	1.815	1.853	1.960	0.085	0.72	0.41	0.34
c9t11 18:2 ⁴	0.354	0.267	0.335	0.317	0.028	0.05	0.54	0.19
t10c12 18:2 ⁴	0.016	0.014	0.014	0.016	0.001	0.90	0.72	0.13
c6,9,12 18:3 ⁵	0.030	0.032	0.032	0.035	0.003	0.29	0.40	0.94
c9,12,15 18:3 ⁴	0.597	0.586	0.574	0.591	0.027	0.91	0.69	0.57
c9t11c15 18:3 ⁴	0.032	0.029	0.031	0.029	0.002	0.36	0.91	0.92
c6,9,12,15 18:4	0.035	0.039	0.044	0.040	0.005	0.97	0.29	0.33
19:0	0.087	0.076	0.087	0.084	0.004	0.13	0.36	0.36
20:0 ⁴	0.119	0.107	0.117	0.114	0.004	0.10	0.57	0.24
c9 20:1 ⁴	0.029	0.028	0.031	0.027	0.001	0.15	0.76	0.36
c11,14 20:2 ⁴	0.030	0.027	0.029	0.029	0.002	0.36	0.74	0.23
c8,11,14 20:3 ⁴	0.073	0.071	0.076	0.085	0.007	0.56	0.16	0.39

Continued

Table 3 (Continued). Milk fat composition (g/100 g of fat) within the first 21 d of lactation of cows receiving supplements of biotin, folic acid, and vitamin B₁₂

Item	Treatment ¹				SEM	P-value ²		
	B ₈ -/B ₉ B ₁₂ -	B ₈ +/B ₉ B ₁₂ -	B ₈ -/B ₉ B ₁₂ +	B ₈ +/B ₉ B ₁₂ +		B ₈	B ₉ B ₁₂	B ₈ × B ₉ B ₁₂
c5,8,11,14 20:4 ⁴	0.157	0.151	0.158	0.172	0.009	0.64	0.22	0.29
c8,11,14,17 20:4 ⁴	0.015	0.015	0.014	0.015	0.002	0.93	0.64	0.51
c5,8,11,14,17 20:5	0.057	0.055	0.055	0.058	0.004	0.92	0.94	0.49
22:0 ⁴	0.042	0.032	0.038	0.035	0.002	0.01	0.86	0.11
c13,16 22:2 ⁴	0.012	0.012	0.013	0.012	0.001	0.88	0.44	0.37
c7,10,13,16 22:4 ⁷	0.022	0.019	0.021	0.022	0.002	0.37	0.61	0.15
c7,10,13,16,19 22:5 ⁴	0.086	0.071	0.075	0.076	0.005	0.16	0.60	0.12
c4,7,10,13,16,19 22:6 ⁴	0.018	0.018	0.017	0.017	0.001	0.99	0.31	0.67
24:0 ^{4,6}	0.031	0.024	0.029	0.028	0.002	0.09	0.77	0.22
Glycerol ^{4,5}	11.500	11.362	11.354	11.452	0.058	0.74	0.63	0.04
Sum								
De novo FA ^{4,5,6,8}	17.110	14.449	14.931	16.627	0.980	0.60	1.00	0.03
C16 ⁴	29.995	26.672	26.830	26.598	0.704	0.06	0.10	0.12
Preformed FA ^{4,9}	39.235	44.387	43.792	42.206	1.441	0.20	0.38	0.02

¹Treatments are defined as follows: B₈-/B₉B₁₂- = 2-mL weekly i.m. injection of saline 0.9% NaCl; B₈+/B₉B₁₂- = 20 mg/d of dietary biotin and 2-mL weekly i.m. injection of saline 0.9% NaCl; B₈-/B₉B₁₂+/ = 2.6 g/d of dietary folic acid and 2-mL weekly i.m. injection of 10 mg of vitamin B₁₂; B₈+/B₉B₁₂+/ = 20 mg/d of dietary biotin, 2.6 g/d of dietary folic acid, and 2-mL weekly i.m. injection of 10 mg/wk of vitamin B₁₂. Treatments were given from d -26 to 21 relative to parturition. Least squares means and SEM are presented.

²Statistical comparisons: B₈ effect (B₈-/B₉B₁₂- plus B₈-/B₉B₁₂+/ vs. B₈+/B₉B₁₂- plus B₈+/B₉B₁₂+/); B₉B₁₂ effect (B₈-/B₉B₁₂- plus B₈-/B₉B₁₂+/ vs. B₈-/B₉B₁₂+/ plus B₈+/B₉B₁₂+/); B₈ × B₉B₁₂ (the interaction effect between each simple effect levels of B₈ and B₉B₁₂; that is, B₈- = B₈-/B₉B₁₂- vs. B₈-/B₉B₁₂+/; B₈+ = B₈+/B₉B₁₂- vs. B₈+/B₉B₁₂+/; B₉B₁₂- = B₈-/B₉B₁₂- vs. B₈+/B₉B₁₂-; B₉B₁₂+/ = B₈-/B₉B₁₂+/ vs. B₈+/B₉B₁₂+/).

³c = *cis*; t = *trans*. FA = fatty acids.

⁴Week effect, *P* < 0.07.

⁵B₈ × week interaction, *P* < 0.09.

⁶B₈ × B₉B₁₂ × week interaction, *P* < 0.08.

⁷B₉B₁₂ × week interaction, *P* < 0.06.

⁸Sum of straight even-chain FA from C6 to C14.

⁹Sum of branched-chain FA (*iso* 13:0, *iso* 14:0, *iso* 15:0, *anteiso* 15:0, *iso* 16:0, *iso* 17:0, and *anteiso* 17:0), odd-chain FA (13:0 and 15:0), and all FA with a chain length greater than 17 C.

compared with B₈-/B₉B₁₂- cows. The increased milk performance was associated with a tendency for greater plasma concentration of nonesterified FA (Duplessis et al., 2022). Even though these cows mobilized more FA from body reserve, this did not translate into excessive plasma BHB concentration (Duplessis et al., 2022), lessening the odds of possible cow health impairments, such as ketosis (van Kneysel et al., 2005; Ospina et al., 2010) and suggesting that the supplementation of B₉B₁₂ enhanced efficiency of energy metabolism in early lactation. Nevertheless, Zachut and Moallem (2017) concluded that cows with greater BW loss (7.6% of BW) within the first 5 wk of lactation had lower reproductive performance than cows with lower BW loss (5.3% of BW) during the same period without an effect on the incidence of ketosis, metritis, and retained placenta. The current experimental design did not allow following cows' reproductive performance beyond 21 DIM. It is therefore not known if the greater BW loss for B₈-/B₉B₁₂+/ could have affected reproductive performance. A previous study suggested that supplementation in B₉B₁₂ given from d -21 to 56 relative to

calving decreased DIM at first breeding (Duplessis et al., 2014a). However, B₉B₁₂+/ cows in that study lost less BW within the first 56 d of lactation (Duplessis et al., 2014b). It is noteworthy that results indicate that the greater body fat mobilization observed for supplemental B₉B₁₂ began after parturition, suggesting that the body fat mobilization was supporting the greater energy demand for milk production. Indeed, prepartum BW change and EB of B₈-/B₉B₁₂+/ cows were not different from those of B₈-/B₉B₁₂- cows, and colostrum fat composition did not indicate lower de novo or greater preformed FA concentrations. No treatment effect was observed on BCS, as was also reported by Graulet et al. (2007), although Duplessis et al. (2014b, 2017b) reported that BCS postpartum was greater for B₉B₁₂+/ cows. Body condition scoring is a subjective evaluation of body tissue reserve and is independent of BW (Sakaguchi, 2009), explaining the conflicting results between postpartum BCS and BW. Cows in the B₈+/B₉B₁₂- group also had a milk FA profile suggesting body fat mobilization after calving (i.e., lower milk de novo and greater preformed concentrations, such as

c9 18:1). Values for milk de novo and preformed FA concentrations were similar between B₈+ / B₉B₁₂- and B₈- / B₉B₁₂+ cows. Nevertheless, supplemental B₈ did not significantly increase milk yield to the extent that supplemental B₉B₁₂ did within the first 21 d of lactation. Milk FA profile of cows receiving B₈ supplement in the experiment of Enjalbert et al. (2008) also suggested greater body fat mobilization in early lactation. However, in that trial, cows receiving dietary B₈ supplement produced 10% more milk than control cows, which was not the case in the current study. Enjalbert et al. (2008) did not report actual DMI between treatments; therefore, we cannot exclude the possibility that the greater lipid mobilization might have been caused by a decrease in DMI for B₈+ cows, as was observed in the current experiment. The effect of B₈ supplement on rumen function, as discussed earlier, might have impeded the milk performance increase in the current study.

Lipid synthesis might be hindered by a lack of B₈, because this vitamin is a coenzyme for acetyl-CoA carboxylase, which has a role in FA elongation (Combs and McClung, 2017). In the mammary gland, this pathway synthesizes de novo FA between 5 and 16 C. However, in the current study, supplying B₈ did not translate into greater de novo FA concentration or yield; in fact, some of them even decreased. This suggests that, under the conditions of the current study, B₈ was not a limiting factor for acetyl-CoA carboxylase in mammary gland tissue.

Results from this study highlight the variability in response following supplementation of B₈, B₉, and B₁₂. To our knowledge, this is the first time that an unprotected B₈ supplement given during the transition period is reported to affect rumen function, possibly through an effect on ruminal microflora. This should be further investigated under different management conditions and for a longer period. Variable responses to B₉B₁₂ supplementation in early lactation have also been reported in the literature (Girard and Duplessis, 2022). Most studies reported that a B₉B₁₂ supplementation increased the efficiency of energy metabolism in early lactation, either by increasing milk yield with similar DMI, BCS, or both (Graulet et al., 2007; Preynat et al., 2009b) or by suggesting a decreased body reserve mobilization with similar milk yield and DMI (Duplessis et al., 2014b, 2017b). To our knowledge, this is the first time that increased body reserve mobilization in B₉B₁₂+ cows has been reported. However, such a great increase in milk yield following B₉B₁₂ supplement has never been previously reported. More research is needed to predict under which circumstances B₉B₁₂ supplementation might be beneficial to cows during the transition period.

CONCLUSIONS

Despite similar postpartum BCS and DMI regardless of B₉B₁₂ supplementation, within the first 21 DIM, we observed a 13.5% increase in milk yield for B₈- / B₉B₁₂+ cows compared with B₈- / B₉B₁₂- cows. However, these cows mobilized more body fat reserve, as suggested by greater BW loss and preformed FA concentrations in milk fat. Cows in B₈+ / B₉B₁₂- also had a milk FA profile indicating body fat mobilization, although milk yield was not increased. Dietary supplemental B₈, which was unprotected from ruminal degradation, probably altered rumen function, as indicated by a decrease in some ruminal biohydrogenation intermediates and odd- and branched-chain FA in milk fat. This could explain the lower postpartum DMI and the absence of increased milk yield for B₈+ cows. Results suggest that, under the current conditions, B₈ was not limiting milk-ing performance of the animals.

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